Universidade Vigo

Evolutionary Computation – Lab-Sessions

Escola Superior de Enxeñaría Informática
Edificio Politécnico
Campus universitario
32004 Ourense

http://esei.uvigo.es
mailto:formella@uvigo.es

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1. First Week

Objectives: Install all necessary packages in python (maybe we will extend the tools in the following weeks). Run the jupyter notebooks to check that everything works. Get some first impressions of how genetic algorithms (and other heuristic algorithms) work on two problems: minimizing a real-valued multi-dimensional function and the traveling salesperson problem (TSP).

1. Download all files ncessary for this week from the web-page http://formella.webs.uvigo.es/doc/ec25/index.html or from the corresponding teaching platform available at your university.

2. Install all necessary python components, including the `scikit-opt` package from https://github.com/guofei9987/scikit-opt

   a) For instance, look at https://docs.anaconda.com/anaconda/install on how to install Python with the Anaconda distribution.

   b) Under a decent Linux system you might proceed as follows (note that roughly **7 GB** of disk space will be used):

   ```
   # navigate to a suitable location in your filesystem, for instance:
   mkdir EC_files
   cd EC_files
   # download anaconda installer and install according to your preferences
   wget https://repo.anaconda.com/archive/Anaconda3-2024.10-1-Linux-x86_64.sh
   bash Anaconda3-2024.10-1-Linux-x86_64.sh
   # download zip-file of source-code from site (maybe same location)
   # https://github.com/guofei9987/scikit-opt
   unzip scikit-opt-master.zip
   cd scikit-opt-master
   pip install .
   # this should install at least the version 0.6.6 from the local directory
   # if you like python-files synchronized with jupyter notebooks
   conda install jupytext
   jupytext --set-formats ipynb,py:percent FirstSession_tsp.ipynb --sync
   # or whatever files you want to synchronize
   ```

   You find `jupytext` documentation at https://jupytext.readthedocs.io/en/latest/

   c) Under Windows or Mac, please refer to the documentation of your system and the packages how to proceed with an installation.

3. Take a look at the web-pages:

   a) https://www.sfu.ca/~ssurjano/optimization.html
   b) https://www.kaggle.com/code/kooaslansefat/cec-2017-benchmark
   c) http://comopt.ifi.uni-heidelberg.de/software/TSPLIB95/

4. Run the 2 jupyter notebooks to check whether everything works fine.

5. Play around with the parameters regarding: objective functions, genetic algorithm parameters, Monte Carlo rounds, underlying datasets, etc.

6. Generate some statistics output, especially mean and variance, for a certain number of Monte Carlo rounds and genetic algorithm iterations for parameter sets that you found interesting.